

Cultivating Digital Agriculture

The Future is in the Cloud



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Academia

- Technische Universität Berlin:
Information Systems
- Birkbeck, University of London:
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Industry

- Fraunhofer FOKUS (Berlin)
- Springer Nature (Berlin/London)
- R3 (London)
- Amazon Web Services (Munich)
- Google (Dublin/Berlin)



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Alphabet

Google Cloud is a part of the Alphabet family of companies



Google Cloud



Consumer
Health
Products



Android
Mobile operating
systems



Ads
Search Engine
Marketing



Google Play
App store



Chrome
Enterprise



Maps
Mapping, location
services & logistics



Search
Search engine



YouTube
Internet video
service

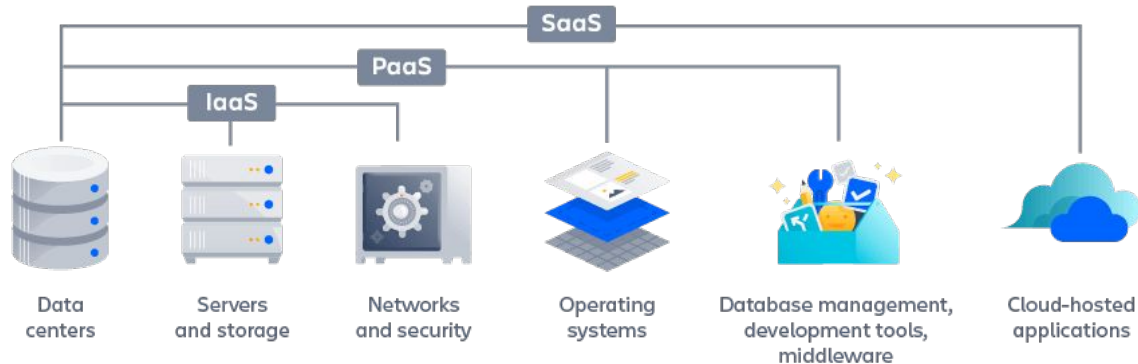


"Biking on the Google campus"
Image Source: **Google**

Cloud Computing

Cloud computing is the on-demand availability of computing resources as services over the internet.

It eliminates the need for enterprises to procure, configure, or manage resources themselves, and they only pay for what they use.



Challenges for Agriculture in Hungary and Beyond

Climate Change (e.g., soil erosion and biodiversity loss) *

Consumers' changing tastes and expectations *

Rising demand for food and food quality

Need for improving farm productivity *

Need to Inspire young people to stay in rural areas

Building resilience against global economic factors *

Source: **Syngenta Group**

(<https://www.syngenta.com/en/innovation-agriculture/challenges-modern-agriculture>)



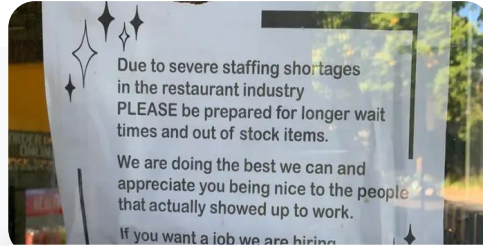


To achieve margin expansion greater operational efficiency is required

The grocery store is arguably where consumers feel inflation most.



45% of the average company's one year earnings will be lost over the next decade due to supply-chain disruption¹



4.5 million workers quit their jobs in November 2021, with the food services and retail trade being the most impacted industries²

Successful Farming

FOOD PRICES RISE AT FASTEST PACE IN 41 YEARS

The New York Times

Food Prices Approach Record Highs, Threatening the World's Poorest

The prices have climbed to their highest level since 2011, according to a UN index. It could cause social unrest "on a widespread scale," one expert said.



ALJAZEERA

Economy | Business and Economy

China's prices surge past forecasts amid lockdowns, Ukraine war

Soaring factory inflation in world's second-largest economy could exacerbate rising prices worldwide.



Bloomberg

Soaring Prices for Everything Used to Make Food Brings More Inflation

Farmers are facing increasingly tough margins at a time when global demand is already tight.

FINANCIAL TIMES

Food Prices (+ 48% in 12m)

Global shoppers feel the pinch of rising food prices

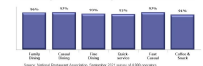


NEWS RELEASE October 25, 2022

Rising food costs and supply chain issues are creating challenges

The restaurant industry is being squeezed from the impact of supply chain issues that are currently impacting many sectors across the economy. In a September 2022 survey conducted by the National Restaurant Association, 57% of respondents said their restaurant experienced supply delays or shortages of key food or beverage items in recent months.

Percent of restaurants that experienced supply delays or shortages of food/beverage items in recent months



Sources:

1.McKinsey Getting ahead of supply chain risk

2.Forbes: The Great Resignation rages on as a record 4.5 million Americans quit, January 4, 2021

New York Times Food Prices Approach Record Highs, Threatening the World's Poorest

Successful Farming Food Prices Rise At Fastest Pace in 41 years

Bloomberg Soaring Prices for Everything Used to Make Food Brings More Inflation

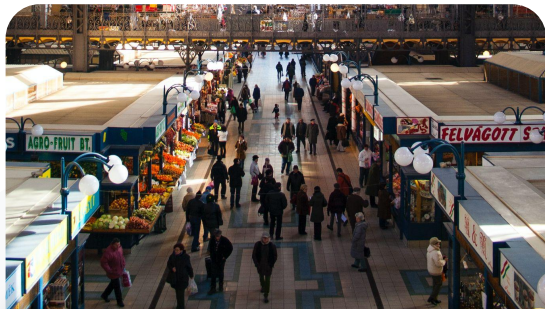
National Restaurant Association Rising food costs and supply chain issues are creating challenges

Al Jazeera China's prices surge past forecasts amid lockdowns, Ukraine war



Shifting consumer diets & channels of trade

Multiple dynamics are impacting how and why people shop



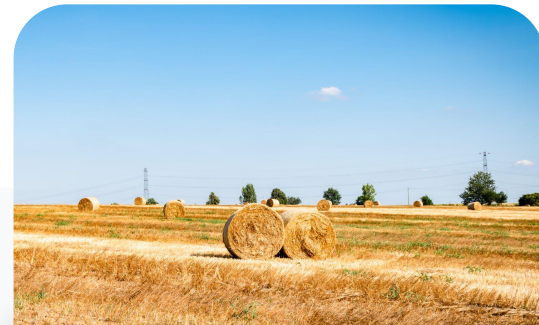
Increase in Omnichannel Commerce

50-70% of consumers intend to continue using alternatives to in-store shopping.¹



Aligning Values

82% of consumers say that sustainability is more top of mind now than it was before COVID-19²



Sustainable Practices

Sustainability-marketed products have delivered 54.7% of CPG market growth.³

1. McKinsey 2022 Global Sentiment Survey

2. Google / C Space qualitative survey activity, US, UK, MX, BR, DE, FR, IN, n = 528, November 24 - 26 2020.

3. NYU Stern Sustainable Market Share Index™



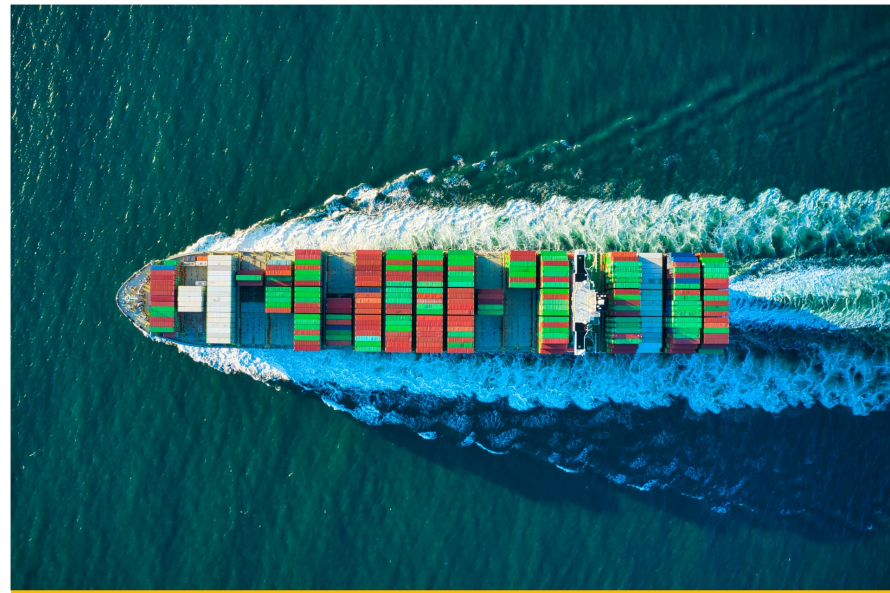
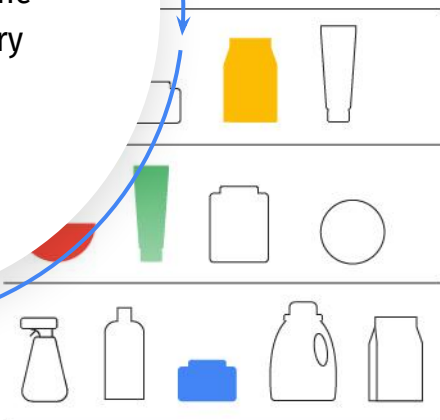
Shifting consumer & commerce trends are forcing retailers to rethink business planning

Out of Stocks cost CPG

\$47B¹

Annually in the US alone¹
and globally inventory
distortion costs

\$1.8T²





Moving from talk to sustainable to actions

The regulatory landscape is increasingly evolving

Since 1972, we've seen:

150

countries have enshrine **environmental protection** or the right to a healthy environment **in their constitutions**

78

nations strengthen **their environmental laws** after adopting a constitutional right to a healthy environment

38x

growth in environmental laws and agencies

Regulatory violations have led to the filing of over **1,305 climate-related legal cases** globally

Evolving environmental laws across countries



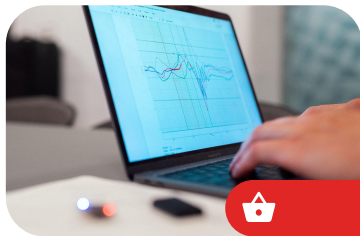
Successful enterprises are breaking down their data silos

Powering business functions with smart insights in real time



Quest for margin expansion

Improve productivity/yield
at every stage of
operations



Shifting consumer diets & channels of trade

Build timely, predict
consumer insights on
health and nutrition trends



Agility and reliance for a risky world

Blend internal data with
external data to enhance
predictions and
decision-making



Moving from talk to action on sustainability

Embed sustainability into
food production key
processes and best
practices



Quality assurance & product defect detection

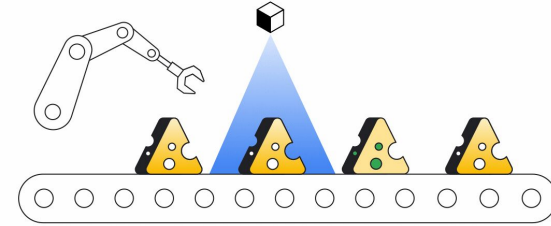
Google Cloud Visual Inspection AI transforms manufacturing quality with top-ranked computer vision and ML capabilities to detect tiniest defects at an unprecedented accuracy and scale

Manufacturing

- Surface defect: scratch, dent, crack, split, color failure, debris
- Process conformance: size, shape, adhesive presence

Visual Inspection

Identify defects in production process using machine vision to improve product quality and customer satisfaction



Assembly

- Conformance: correct part, position, alignment and sequence
- Defect: missing part, incorrect specification, unaligned, debris

**Value
Opportunity**



Top-line
growth



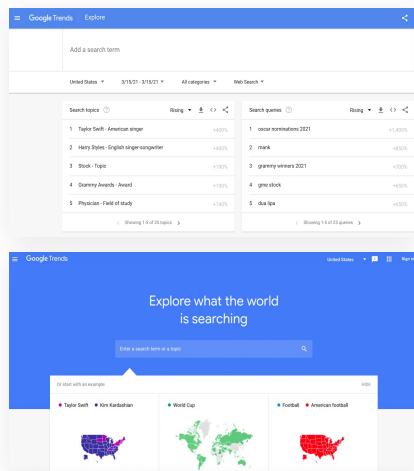
Risk
reduction



Consumer-backed predictive trends & insights

Amplify insights with unique Data Signals from Google Trends directly into BigQuery

Google Trends



Google Data Signals



BigQuery
Datasets

Data Signals UI



Customer's
Project

*still in development,
limited to the US*



Marketing

What searches are trending that should impact my campaign planning?



Merchandising & Demand Forecasting

How can I understand demand trends to make better assortment planning decisions?



Supply Chain

What trends might drive higher sell through or OOS?



Executives

What consumer search trends might impact business planning?

**Value
Opportunity**

**Top-line
growth**

**Speed of
innovation**



Inventory Optimization

Respond to risk and disruptions by optimizing your inventory parameters and collaborating with your partners

Inventory Optimization AI

ALERT:



(possible demand/supply mismatch due to weather event)



**Dynamic resetting of
inventory parameters**

Google Cloud Supply Chain Pulse



Real-time visibility and analytics



Alert-based event management



Collaboration across teams and partners

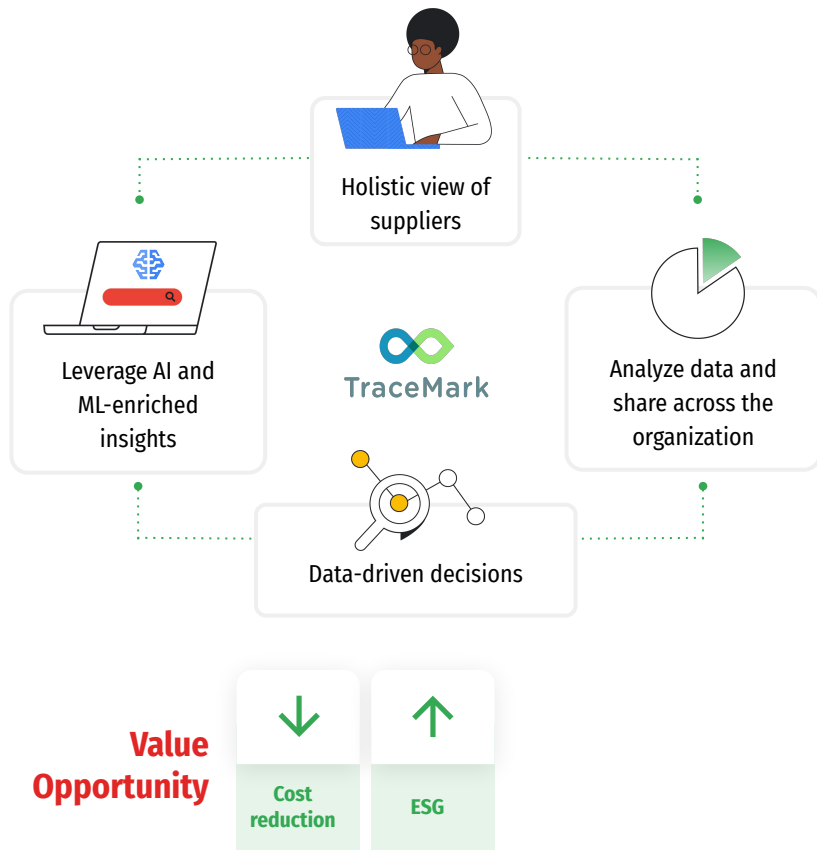


Ready to go BI templates

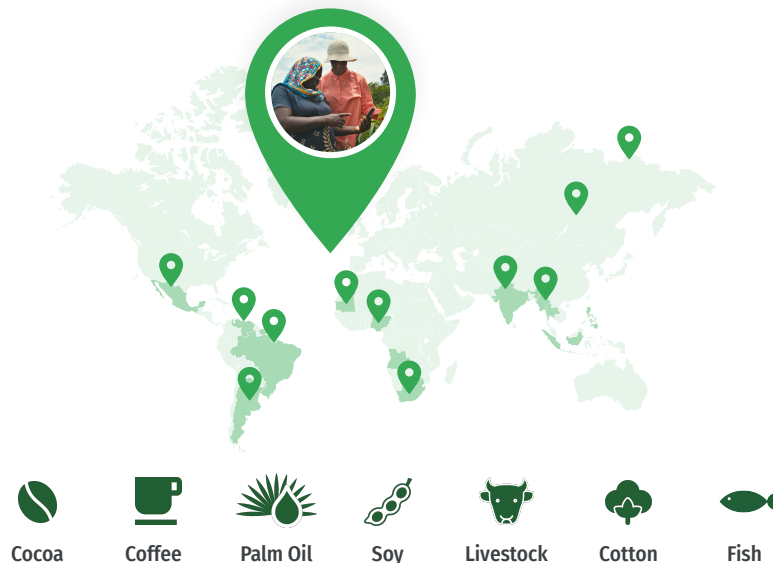


Sustainable Sourcing

Google Earth Engine + TraceMark empower CPGs with a deeper understanding of raw material sourcing practices across supplier networks



The most productive crops are in high demand and are facing increased supplier scrutiny



Our work today aligns to industry business objectives

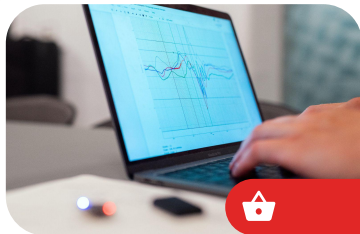


Quest for margin expansion

Quality assurance &
product defect detection

Predictive maintenance &
OEE management

Commodity price risk
management



Shifting consumer diets & channels of trade

Consumer-backed
predictive trends &
insights



Agility and reliance for a risky world

Inventory optimization

Customer Risk Delivery
Prediction

Supply forecasting

Logistics fleet optimization



Moving from talk to action on sustainability

Machine Learning

Waste-reducing and energy
efficient production

Case Study



Case Study: *Prediction of Climate Hazards*

Hazard	Impact
Less rain (Dry spell)	Submergence
	Poor grain set and grain development
	Anaerobic condition rotting
Unseasonal rain (Heavy rains)	Pre-harvest sprouting
	Workability
Wind	Lodging
	More time and cost to harvest

Source: Shah, H., Hellegers, P., and Siderius, C. (2021). Climate risk to agriculture: A synthesis to define different types of critical moments. In *Climate Risk Management* (Vol. 34). Elsevier, 100378. <http://dx.doi.org/10.1016/j.crm.2021.100378>

Preparedness

It would be very helpful if we knew where risks for certain crops would occur in future.

- **Mitigate risk** (introduce irrigation, windbreaks, etc.)
- **Circumvent risk** (apply different crops)
- **Manage risk** (insure risk-prone areas differently)
- **Avoid risk** (do not cultivate)



Hypothesis 1

Hints on climate change hazards are present in satellite images, even though we might not be able to notice them.

Risk of anaerobic
condition rotting?



Risk of submergence?



Risk of lodging
due to winds?



Big Data



Image Source: Josef Steppan (via Wikipedia)

Little data → Easy problems
(MNIST: 60,000 handwritten digits)



Image Source: OpenCV.org

Much data → Hard problems
(ImageNet Challenge: 1.2 mio images)

Risk of submergence?

Much data!

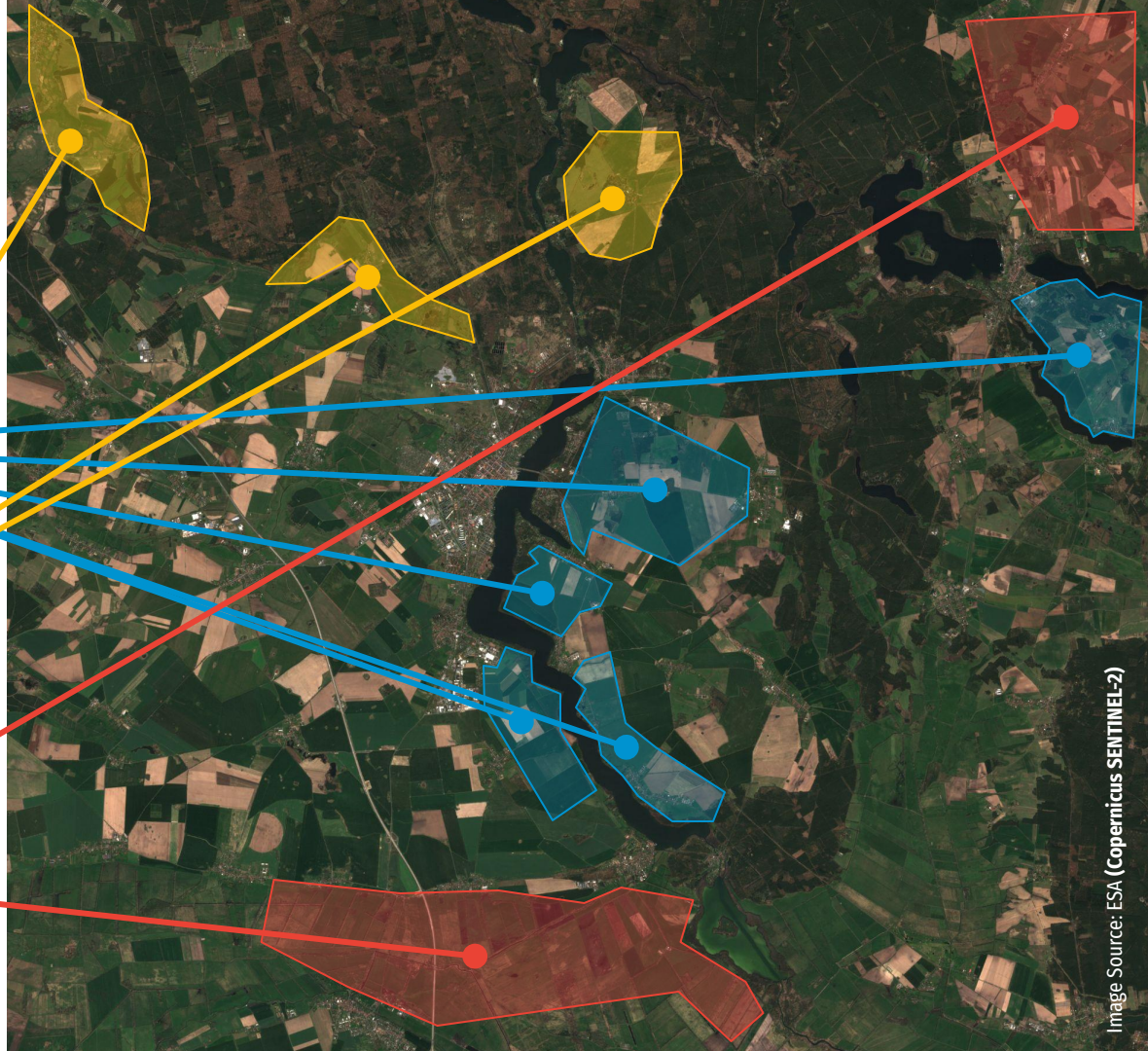
Hazard Data

We might be able to obtain **ground truth** from other regions or points in time.

Areas with anaerobic
condition rotting

Areas with occurrences
of submergence

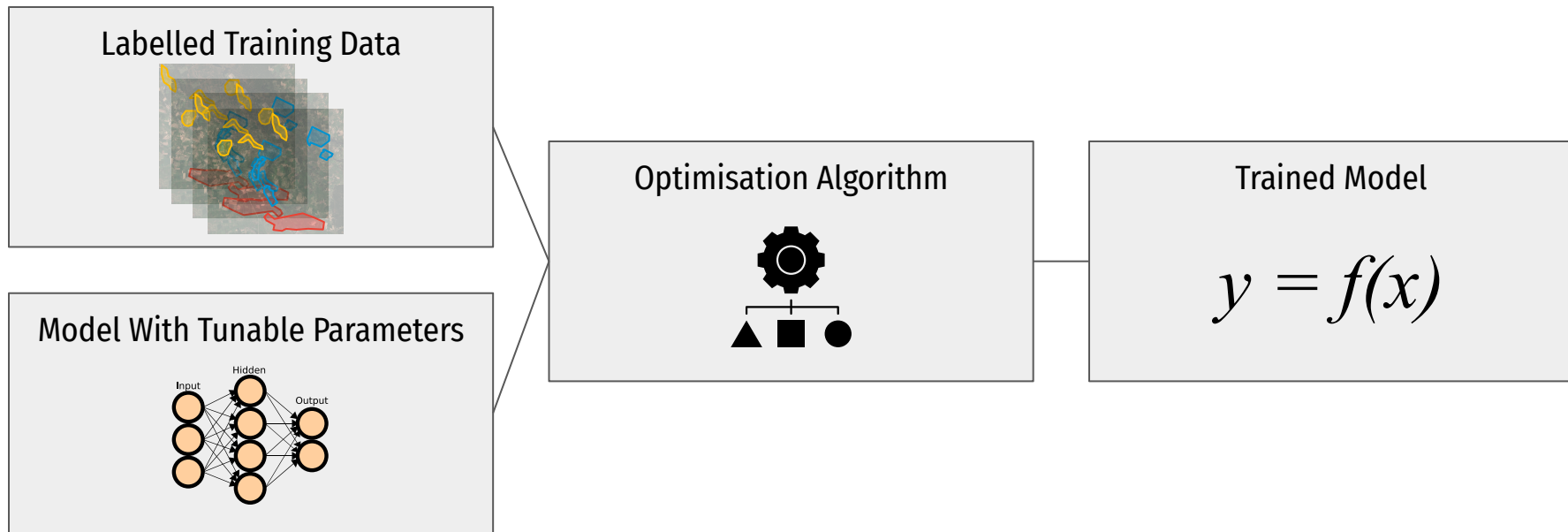
Areas with significant
observed lodging



Machine Learning to the Rescue?

“The goal of machine learning is to come up with algorithms that can learn how to perform a certain task based on example data.”

<https://medium.com/onfido-tech/machine-learning-101-be2e0a86c96a>



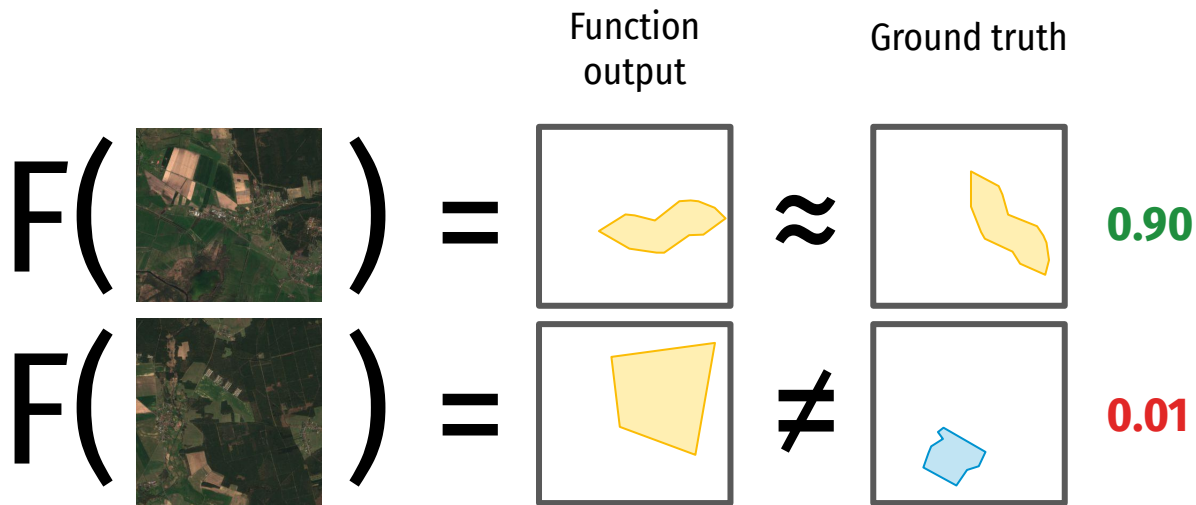
Training process

1. Prepare training data:

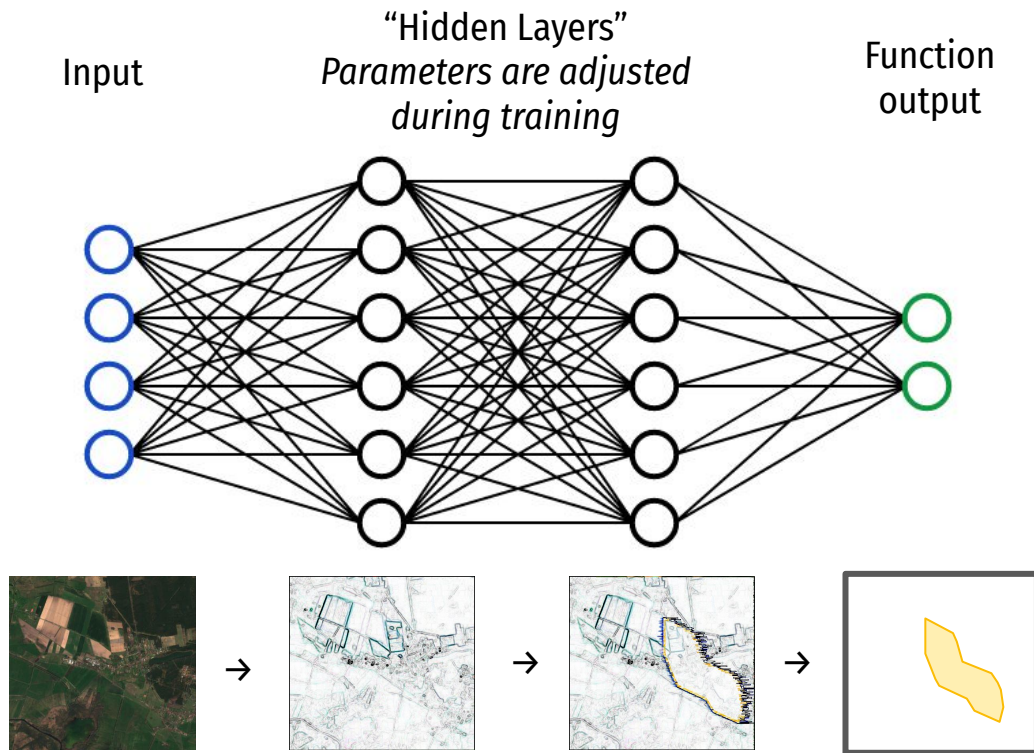


2. Fit a function:

$$F: \{image\} \rightarrow \{shape\}$$

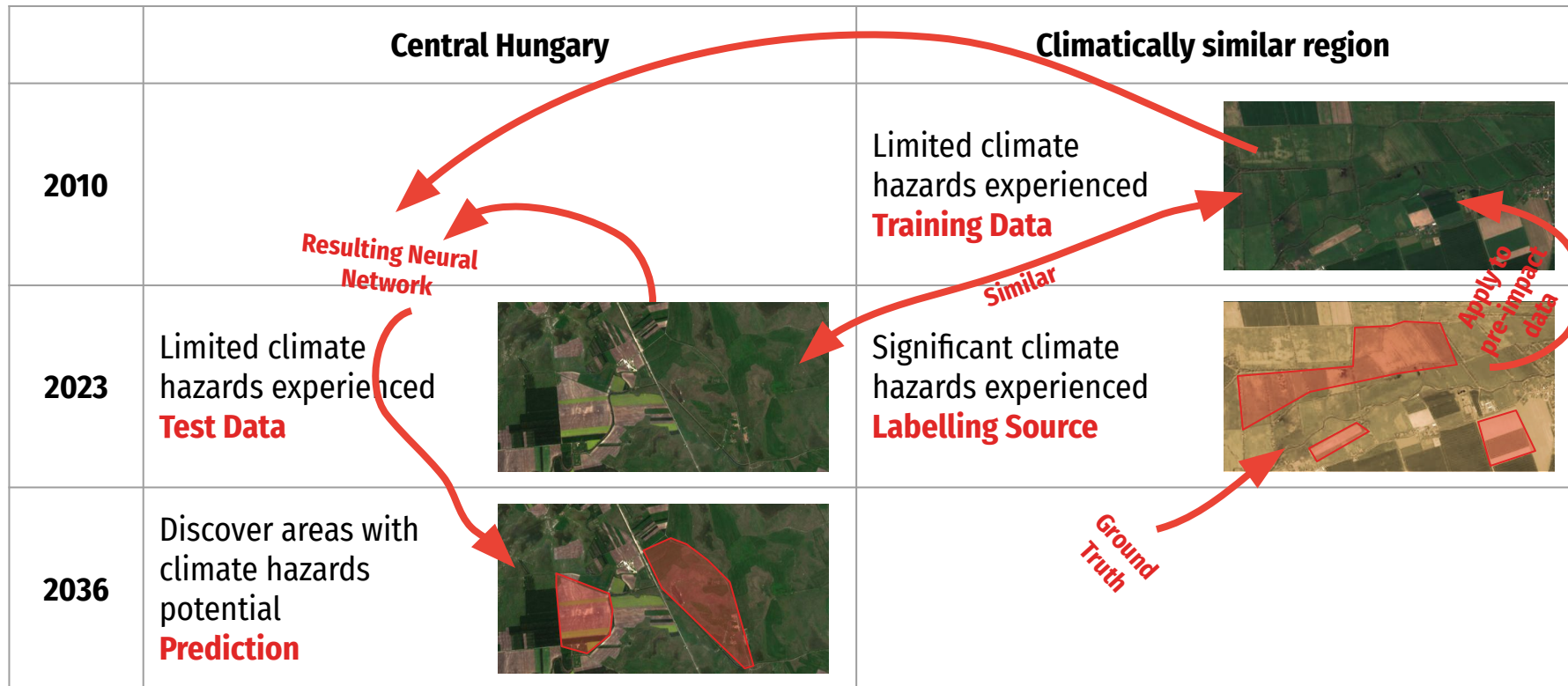


What is F ? Neural Networks!



Hypothesis 2

What is happening in other regions now is a window into the future for Central Europe.



Domain adaptation

Training on **Dataset A** → Testing on **Dataset B**

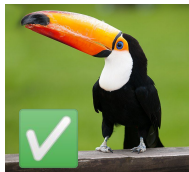
Works if:

1. Datasets are **similar** enough
2. Dataset A was **diverse** enough

✓ Example



Bird/No Bird



*Output:
Bird*

✗ Example



Wolf

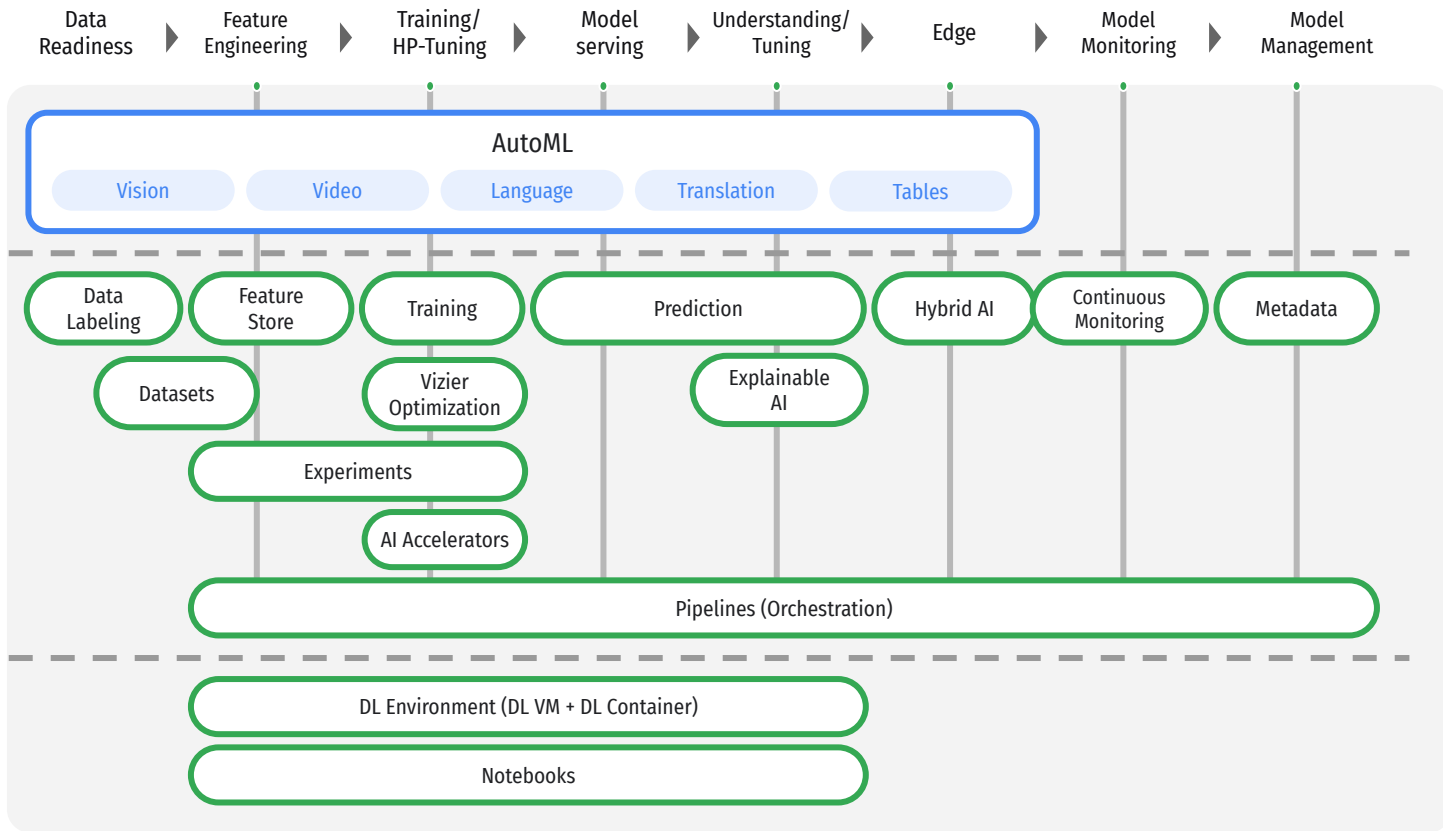
Husky

Wolf



*Output: Wolf
Ground truth: Husky*

Machine Learning on Cloud



Take Home Messages

Key risks create **great opportunities** for those who...

1. ...circumvent and mitigate climate hazards.
2. ...address consumers changing expectations.
3. ...meet the rising demand for high-quality food.
4. ...are resilient against global economic factors.

Cloud computing can **contribute to your success** by using data to...

1. ...detect defects.
2. ...manage commodity price risks.
3. ...predict consumer behaviour.
4. ...optimize inventory.
5. ...predict climate hazards.



Thank You!